

Student Research Internship

Machine Learning Engineer

AquaSentinel AI | Water Quality Prediction for Swiss Rivers & Lakes

About the Project

AquaSentinel AI is an open-source research project building a universal AI layer that predicts dissolved oxygen (DO) in Swiss rivers and lakes — the variable that directly determines whether fish survive summer heatwaves. We are extending AareML, a seq2seq LSTM developed as part of a CAS in Advanced Machine Learning at the University of Bern, which achieved RMSE 0.303 mg/L and zero-shot transfer to 11 unseen Swiss rivers.

The project is funded by the Prototype Fund Switzerland (CHF 50,000) and runs November 2026 – February 2027, with a pilot deployment in collaboration with a Swiss fishery association.

Your Role

You will be the lead ML engineer for the AI prediction component — building, training, and validating the universal DO forecasting model. Work is structured in four monthly milestones:

- **Month 1** — Baseline model: Extend the existing AareML LSTM to predict DO at all available FOEN stations using temperature, flow, and water level data. Establish a reproducible training pipeline.
- **Month 2** — Expanded model: Integrate Eawag Urban Water Observatory and cantonal monitoring networks. Add lake coverage alongside rivers. Evaluate transfer performance across water body types.
- **Months 3–4** — Live integration & validation: Ingest the real-time sensor stream from the IoT team deploying sensors at the pilot river site. Validate DO predictions against ground-truth measurements. Iterate and refine. Produce a documented, deployable model.

What You Will Work With

- Model architecture: seq2seq LSTM (PyTorch), building on the AareML open-source codebase
- Data: FOEN hydrodaten API, CAMELS-CH-Chem (40 years, 115 catchments), Eawag open datasets, cantonal portals
- Infrastructure: UBELIX HPC cluster (Uni Bern), Python, GitHub
- Team: weekly check-ins with project leads Olga and Tatyana; collaboration with the IoT team at OST Rapperswil

What We Are Looking For

Required:

- Enrolled in a BSc or MSc program in Computer Science, Data Science, Environmental Science, or a related field
- Solid Python skills and experience with PyTorch or TensorFlow
- Familiarity with time-series data and ML model training
- Comfortable working independently with real-world, messy datasets

Nice to have:

- Experience with hydrology or environmental datasets
- Knowledge of NeuralHydrology or similar hydrological ML frameworks
- Interest in open-source development and responsible AI

What We Offer

- CHF 1,000/month stipend — 4-month duration (~15–20 hours/week)
- Academic credit: we will work with your university to support a thesis or project course connection
- Co-authorship on any publications arising from the work
- Hands-on experience with a production-oriented open-source codebase and real sensor data
- A working prototype in the field by May 2027 — you will see your model deployed

Logistics

- Duration: 4 months — November 2026 to February 2027
- Location: Remote-first with occasional meetings in Zurich / Bern
- Start date: Flexible, ideally 1 November 2026
- Open to BSc and MSc students from any Swiss or international university

How to Apply

Send a short email (one page maximum) to polar.bear.before.lunch@gmail.com with:

1. Your current program and year of study
2. Two or three sentences on why this project interests you
3. A link to your GitHub or any relevant code or project

We will respond to all applications within one week.